

Agile Software Engineering and DevOps – Lab 5

Title

Designing Test Cases for Different Software Testing Categories

Lab Objectives

By the end of this lab, students will be able to:

1. Understand the **purpose of different testing categories**
 2. Design **effective and meaningful test cases**
 3. Derive test cases from:
 - SRS
 - Use Case Diagrams
 - Sequence Diagrams
 4. Apply **Sanity, Security, Stress, and Performance testing** concepts
 5. Appreciate testing as a **quality assurance activity**, not just defect finding
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Pre-Requisite

- Lab 1 – Waterfall Model
 - Lab 2 – Scrum
 - Lab 3 – SRS
 - Lab 4 – UML (Use Case & Sequence Diagrams)
 - Same **team and same project**
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Testing Categories Covered

Each team must write **2–5 test cases for each category**:

1. **Sanity Testing**
2. **Security Testing**
3. **Stress Testing**
4. **Performance Testing**

Total expected test cases per team: **Around 15.**

Part A – Identify Test Scenarios

Task A1: Identify Core Scenarios

From SRS & UML, identify:

- Critical functionalities
- High-risk areas
- Frequently used features

Examples (Library Management System):

- Login
- Issue Book
- Return Book
- Search Book
- Report Generation

Deliverable:

- List of selected test scenarios
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Part B – Write Test Cases by Category

Teams must write **structured test cases** using the standard template below.

1. Sanity Testing

Purpose

To verify that **critical functionalities work after a build or change**.

Guidelines:

- Focus on *basic workflow*
- No deep validation

Write about 5 Sanity Test Cases

2. Security Testing

Purpose

To ensure the system **protects data and prevents unauthorized access**.

Guidelines:

- Authentication

- Authorization
- Input validation
- Data protection

Write about 5 Security Test Cases

3. Stress Testing

Purpose

To evaluate system behaviour under **extreme load conditions**.

Guidelines:

- High number of users
- Large data volumes
- Limited resources

Write 4 Stress Test Cases

4. Performance Testing

Purpose

To verify **response time, throughput, and resource usage**.

Guidelines:

- Response time
- Concurrent users
- System scalability

Write 4 Performance Test Cases

Part C – Review & Reflection

Task C1: Validation Questions

Each team answers:

1. Are all test cases traceable to SRS/use cases?
 2. Which testing category was hardest to write? Why?
 3. Which defects would **not** be caught by sanity testing?
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STANDARD TEST CASE TEMPLATE

Students should use the same template for all test cases.

Field	Description
Test Case ID	Unique identifier
Test Category	Sanity / Security / Stress / Performance
Test Scenario	What is being tested
Preconditions	System state before execution
Test Steps	Step-by-step actions
Test Data	Input values
Expected Result	Expected system behavior
Actual Result	(Leave blank in lab)
Status	Pass / Fail (Leave blank)

SAMPLE TEST CASES – Library Management System

(You may share these as **reference only**, not for copying.)

1. Sanity Testing – Sample

TC-SAN-01

- Scenario: Verify user login
- Preconditions: User is registered
- Steps:
 1. Open login page
 2. Enter valid username and password
 3. Click Login
- Expected Result: User is logged in successfully

TC-SAN-02

- Scenario: Issue book
 - Preconditions: Book is available, user logged in
 - Expected Result: Book is issued and reflected in user account
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2. Security Testing – Sample

TC-SEC-01

- Scenario: Login with invalid credentials
 - Test Data: Incorrect password
 - Expected Result: Access denied with error message
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TC-SEC-02

- Scenario: Unauthorized access to admin page
 - Preconditions: User logged in as Member
 - Expected Result: Access denied
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3. Stress Testing – Sample

TC-STR-01

- Scenario: Concurrent login attempts
 - Load: 500 users simultaneously
 - Expected Result: System remains stable without crashing
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TC-STR-02

- Scenario: Issue books until inventory limit
 - Expected Result: System handles max transactions correctly
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4. Performance Testing – Sample

TC-PERF-01

- Scenario: Search book response time
 - Load: 100 concurrent users
 - Expected Result: Response time < 2 seconds
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TC-PERF-02

- Scenario: Login response time
- Expected Result: Login completes within acceptable time